

THUSHANGHI SURESHKANTHA

CONTACT

+94 785366645

thushanghisuresh33@gmail.com

No: 14, Janaki Lane, Colombo 04

www.linkedin.com/in/thushanghi-sureshkantha-b895a32a6

HARD SKILLS

- Medical Instrumentation
- Medical Device & Equipment Lifecycle
- Robotics
- Research and development - Arduino,
- Electrical & Electronic Principle - Multisim
- Engineering Mathematics
- Computer Vision - MATLAB, Python
- Internet of Things - TinkerCad, Proteus
- Artificial Engineering & Machine Learning
- Wireless Networks - Cisco
- Programming for Engineers - Python, C++, C#, Python, Java, PHP, HTML, CSS, JavaScript, React, Node.js, MYSQL, XML, NOSQL
- Engineering Science
- Digital Principles
- Engineering Design - SkechUp
- Anatomy & Physiology
- Polymer Materials
- Nanomaterials & Their Technology

PROFILE

Biomedical Engineer Graduate and Biomedical Engineer Trainee at SIBA Holdings, currently receiving hands on training at Kings Hospital Colombo, with a focus on medical robotics, robotic surgical systems, and healthcare technology. Experienced in learning and supporting medical robotics, robotic surgical systems, automation technologies, and robotic-assisted healthcare solutions within a hospital environment. Currently pursuing a Bachelor's degree in Information Technology (BIT), combining biomedical engineering knowledge with IT and technology skills to contribute to the development, operation, maintenance, and advancement of modern medical technologies.

EDUCATION

BEng (Hons) in Biomedical Engineering (Top Up) - (First Class Honours) JUN 2025 – JUN 2026
London Metropolitan University (UK) – Esoft Uni

Bachelor of Information Technology (BIT) OCT 2024 – PRESENT
University of Colombo School of Computing(UCSC - UOC)

Higher National Diploma in Biomedical Engineering - (Merit) NOV 2023 – JUN 2025
Pearson – Esoft Uni

Level 3 Diploma in Information Technology OCT 2023 - OCT 2024
Esoft Uni

Foundation in Information Technology FEB 2024 – AUG 2024
University of Colombo School of Computing(UCSC - UOC)

G.C.E. Ordinary Level JUNE 2022 (2023)
C. Ramanathan Hindu Ladies' College, Colombo - 04
8A, 1B
Mathematics - A , English - A, IT - A, English Literature - A,
Business & Accounting - A, Science - B, Tamil - A, Hinduism - A, History - A

EXPERIENCE

Trainee Biomedical Engineer AUG 2026 - PRESENT
Siba Holdings - | Kings Hospital Colombo

- Receiving practical training in Biomedical Engineering and Medical Technology within the hospital environment.
- Gaining hands-on exposure to surgical robotic systems, including the SSI Mantra Surgical Robotic System used for soft tissue surgery.
- Learning the setup, operation, docking/undocking, and basic troubleshooting of surgical robotic systems in the Operating Theatre (OT).
- Developing practical knowledge of robotic system components, medical imaging systems, endoscopy equipment, and associated electrical and electronic systems.
- Gaining experience in preventive maintenance, functional testing, calibration, and safety checks of medical equipment.
- Working alongside biomedical engineers and clinical staff to understand hospital equipment management and clinical workflows.

SOFT SKILLS

- Analytical thinking & problem solving
- Research & critical analysis
- Attention to detail
- Teamwork & collaboration in Multidisciplinary environments
- Communication & technical presentation skills
- Adaptability & quick learning of new technologies
- Time management & organization
- Creativity & Innovation in healthcare solutions

LANGUAGES

- English (Fluent)
- Sinhala (Intermidiate)
- Tamil (Fluent)

REFERENCE

Mr. Zakir Saleem
Lecturer and Biomedical Engineer
Phone: +94 77 653 8016
Email : zakir.saleem@esu.lk

Mr. Ethiraj Thipakaran
Lecturer and Biomedical Engineer
Phone: +94 78 279 2821
Email : ethiraj.thipakaran@esoft.lk

PROJECTS

Smart Health Monitoring Pillow — APR 2025 – PRESENT
Arduino IDE | C++ | Supabase | React
- Developed a smart pillow using ESP32 to monitor temperature, pulse rate, and breathing patterns.
- Integrated real time data transmission with Supabase and built a React dashboard for visualization.
- Designed a non invasive, cost-effective system for sleep monitoring and apnea detection.

IoT based Health Monitoring System with Blynk — FEB 2026– MAY 2026
ESP32 | MAX30102 | DS18B20 | DHT22 | Blynk IoT
- Developed an IoT based health monitoring system using an ESP32, MAX30102, DS18B20, and DHT22 sensors to collect real time heart rate and temperature data.
- Programmed the system using Arduino IDE and transmitted sensor data via Wifi to the Blynk IoT cloud platform for remote monitoring.
- Implemented real time data visualization and alert functionality using Blynk and a buzzer.

Smart Medication Reminder System — MAR 2026 – MAY 2026
ESP32 | DS1307 RTC | 7-Segment Display
- Developed an IoT based smart medication reminder system using an ESP32 microcontroller.
- Integrated a DS1307 RTC, 7 segment display, LEDs, buzzer, and push button for medication scheduling and alerts.
- Programmed the system in Arduino IDE using Blynk, RTClib, and Wifi libraries.
- Enabled remote medication scheduling, pill count monitoring, and notification confirmation through the Blynk smartphone app.

Sleep Apnea Detection using AI & Machine Learning — OCT 2025- DEC 2025
Python | Machine Learning | Data Analysis
- Developed an AI-based system to detect sleep apnea using physiological data.
- Applied preprocessing and trained models including Logistic Regression, Random Forest, XGBoost, and ANN.
- Evaluated performance using accuracy, ROC-AUC, and visual comparisons to identify the best model.

Spider Robot — MAY 2025 – JUNE 2025
C++ (Arduino IDE)
- Designed and programmed a spider robot using Arduino Nano, MG90 servo motors, and HC06 Bluetooth communication for coordinated movement.
- Integrated servo motors and sensors for obstacle detection and stability.
- Developed control algorithms for multi-leg synchronization and navigation.

Medical Equipment Management System — OCT 2024 – DEC 2024
Python (PyCharm IDE)
- Developed a system to manage medical inventory, orders, and customer records.
- Integrated database functionality for secure data storage and retrieval.
- Improved workflow efficiency through automation and user-friendly interface design.

Sanitation Disinfection Gate Models — FEB 2024 – JULY 2024
SketchUp
- Developed a system to manage medical inventory, orders, and customer records.
- Integrated database functionality for secure data storage and retrieval.
- Improved workflow efficiency through automation and user friendly interface design.

Login and Registration System — SEP 2024 – OCT 2024
C# | SQL Server (Windows Forms)
- Built a secure user authentication system using C# and SQL Server.
- Implemented password hashing and access control for enhanced security.
- Designed an intuitive interface for seamless user registration and login.

OTHER COURSES

- AI & ML Engineer - SLIIT
- Full Stack Development - Moratuwa
- Arduino and programming in IoT - Alison
- ISO 13485:2016 (Medical Device Quality Management System) - Alison
- Diploma in Electrical studies - Alison